The Universal Ghost Clock: repulsion, ride caps, and the census budget

with a verified import audit of arXiv:2603.11066

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Abstract

We generalize the two-clock structure of the companion note to *every* periodic ghost at once: for any valuation signature $\sigma = (k_1, \ldots, k_\ell)$ of depth $K = \sum k_i$, the phantom root $\rho_\sigma = C_\sigma/(2^K - 3^\ell)$ carries an exact clock — alignment $a(x) = v_2(x - \rho_\sigma)$ drops by exactly K per ridden block, the ride lasts exactly $\lfloor (a-1)/K \rfloor$ blocks, and positivity caps $a \le \log_2(|2^K - 3^\ell|\, x + C_\sigma)$. The -1 and $+1$ clocks are the $K = 1, 2$ instances; the $(1,1,2)$-channel ghost $-19/11$ of the Target-2 analysis is the $K = 4$ instance, which resolves the question left open there: *the parity channel is metered exactly like the -1 ghost.* The without-replacement depth supply extends verbatim to every phantom root. These statements were prompted by the manuscript arXiv:2603.11066v6 (Chang), whose relevant pieces we audited: the repulsion mechanism, the census counting formula (after restoring a missing hypothesis), and the amortized census constant $R \le 0.0893$ all *verify* (we recompute $R = 0.08824$ independently); its burst–gap calculus is an exact relabeling of our two clocks (its Persistent Exit Lemma is our parity handoff at $h = 3$); but its shadow return-time theorem is *false as stated* — re-entry gaps of 1 occur, and 2-adic alignment can jump $+16$ bits in one step — and must be replaced by the supply bound.

1 Setup

A *signature* is a word $\sigma = (k_1, \ldots, k_\ell)$, $k_i \ge 1$, of depth $K = \sum k_i$; write $B_j = k_1 + \cdots + k_j$. The block map is $F_\sigma(x) = (3^\ell x + C_\sigma)/2^K$ with $C_\sigma = \sum_{j=0}^{\ell-1} 3^{\ell-1-j} 2^{B_j}$, and its unique fixed point is the *phantom root*

$$\rho_\sigma = \frac{C_\sigma}{2^K - 3^\ell} \in \mathbb{Z}_2,$$

a 2-adic unit (C_σ odd, $2^K - 3^\ell$ odd). Define the *alignment* $a(x) = v_2\big((2^K - 3^\ell)x - C_\sigma\big) = v_2(x - \rho_\sigma)$. Our catalogue of ghosts is exactly the family of phantom roots:

σ	ℓ	K	ρ_σ	role
(1)	1	1	-1	the -1 clock (ascent meter)
(2)	1	2	$+1$	the $+1$ clock (trivial cycle)
$(1,2)$	2	3	-5	negative cycle
$(1,1,2)$	3	4	$-19/11$	the parity channel of Target 2
$(1,1,1,2,1,1,4)$	7	11	-17	negative cycle

2 The universal clock

All verification counts below refer to `search/ghostclock.py`, run over the five catalogued families plus 20 random signatures.

Lemma 1 (Cylinder threshold)**.** *$a(x) \geq K+1$ if and only if the next ℓ letters of x are exactly σ; the modulus 2^{K+1} is minimal. (Verified: minimal forcing modulus $= K+1$ for all 25 signatures.)*

Theorem 1 (Universal ghost clock)**.** *Let $a = a(x) \geq K+1$. Then:*

*(i) (*Repulsion.*) After one ridden block, $F_\sigma(x) - \rho_\sigma = \frac{3^\ell}{2^K}(x - \rho_\sigma)$, so the alignment drops by exactly K. (Verified: 200 block applications, drop $= K$ exactly.)*

*(ii) (*Ride cap.*) The orbit rides exactly $\lfloor (a-1)/K \rfloor$ consecutive σ-blocks and then deviates. (Verified: 625 cases, equality in all.)*

*(iii) (*Height cap.*) For positive integers x, $2^a \leq |(2^K - 3^\ell)x - C_\sigma|$, so $a \leq \log_2\big(|2^K - 3^\ell|\, x + C_\sigma\big)$: deep alignment costs height, uniformly in σ.*

Proof. (i) is one line from the affine form of F_σ. For (ii): each ridden block requires alignment $\geq K+1$ at its start (Lemma 1) and costs exactly K, so the number of ridden blocks is $\#\{j \geq 0 : a - jK \geq K+1\} = \lfloor (a-1)/K \rfloor$; if the block were followed at alignment $< K$, (i) would force negative valuation of a nonzero integer, and at alignment exactly K the deviating bit sits precisely where the final letter's valuation is read, so the last letter strictly exceeds k_ℓ. (iii) is positivity of the nonzero integer $(2^K - 3^\ell)x - C_\sigma$. □

Corollary 1 (The parity channel is metered)**.** *The $(1,1,2)^\infty$ channel — natural depth law, pure parity bias, the configuration shown in the companion note to evade the g-law alone — is governed by the $K=4$ clock of $\rho = -19/11$: riding j blocks requires alignment $\geq 4j+1$, alignment is repelled at 4 bits per block, and alignment a at height x requires $11x + 19 \geq 2^a$. The question left open in the two-clock note is resolved:* every *ghost channel, not just -1, carries an exact run-length cap.*

Lemma 2 (Universal depth supply)**.** *The points of $[2^k, 2^{k+1})$ with alignment $\geq g$ to ρ_σ form a single residue class modulo 2^g, of cardinality $2^{k-g} \pm 1$; an acyclic orbit consumes them without replacement. Every result of the companion note's supply section (lifetime budgets, upcrossing bound) holds verbatim for every phantom root.*

3 The census budget (imported and verified)

Lemma 3 (Affine census; corrected hypothesis)**.** *Let $x_u = A + Bu$, $\beta = v_2(A - \rho_\sigma)$, $\gamma = v_2(B)$, and suppose $\gamma \leq \beta$. Then for $t > \gamma$,*

$$\#\{0 \leq u < 2^L : v_2(x_u - \rho_\sigma) \geq t\} = 2^{L-t+\gamma}.$$

(Verified: 400/400 random cases.) The hypothesis $\gamma \leq \beta$ cannot be dropped: for $\beta < \gamma$ every x_u has alignment exactly β and the count for $t > \beta$ is 0, not $2^{L-t+\beta}$ (explicit counterexample: $\beta = 4$, $\gamma = 7$, $t = 5$, count 0 vs. 512).

Proposition 1 (Amortized census constant; independent recomputation)**.** *Let $M(K,\ell)$ be the number of primitive necklaces of compositions of K into ℓ parts and*

$$R = \sum_{K=3}^{500} 2^{-K} \sum_{\ell > K/\log_2 3} M(K,\ell)\Big(\log_2 3 - \frac{K}{\ell}\Big).$$

Our computation gives $R = 0.088236$, confirming the bound $R \le 0.0893$ of arXiv:2603.11066 (the tail beyond $K = 500$ is negligible by the Chernoff envelope). Against the contraction budget $2 - \log_2 3 = 0.415$ bits per letter this is a $4.70\times$ margin.

Remark. *In our language: the $K \le 2$ families* are *the two clocks — the renewal baseline. Proposition 1 says that all deeper ghost channels combined, weighted by shadow density 2^{-K} and per-letter gain, supply at most 0.088 bits of drift advantage: an* amortized *complement to the per-family supply bounds of Lemma 2. The orbit-level application (that a single orbit cannot beat this amortization) is precisely the calibration wall; the structural sum itself is unconditional arithmetic.*

4 The burst–gap calculus is the two clocks

The manuscript's coordinates translate exactly onto ours: its run length $k = v_2(n+1)$ is our -1 depth g; its "bursts" are our 1-runs, its "gaps" our descent blocks; its modular laws ($k \sim 2^{-(j-1)}$, mean 3; gap $\sim$Geometric($\frac{1}{2}$), mean 2) are our natural meter laws; its mean-gap threshold $g_* \approx 1.71$ is our crossing constant $1/\log_2(3/2) = 1.7095$; and its two open hypotheses (mean gap $\ge g_*$; mean burst ≤ 2) are literally the two channels of our joint calibration conjecture.

Proposition 2 (Their Persistent Exit Lemma is the parity handoff)**.** *For $x \equiv 27 \pmod{32}$ (depth-2 run end in the persistent class): the run ends in one b=1 step at y with $h = v_2(y-1) = 3$* exactly*, whence by the $+1$ clock a single b=2 letter followed immediately by a new 1-run — "gap length exactly one." (Verified: 4000/4000.) Proof: $x = 32s + 27$ gives $y = 48s + 41$, $y - 1 = 8(6s+5)$ with $6s+5$ odd, so $h = 3$; h odd is precisely the odd-parity handoff of the companion note.*

Its compatible-class census also reconciles: enumerating odd residues modulo 2^{2D+3} whose first D letters lie in $\{1,2\}$ gives exactly $2^{2D+2}(3/4)^D$ classes (verified $D \le 8$), i.e. density $(3/4)^D$ — their $|C_D| = 2 \cdot 3^{D-1}$ is the same density in a coarser modulus convention.

5 Refutation: the shadow return-time theorem

The manuscript asserts (its Thm. on shadow return times) that after an orbit exits the depth-K shadow of a family of length ℓ, it cannot re-enter for $\ell + K - 1$ steps, the proof resting on the claim that 2-adic alignment recovers at most one bit per step. Both fail:

- Measured over 400 random orbits per family, with visits counted ride-aware (a maximal run of consecutive ridden blocks is one visit): minimum exit-to-re-entry gap = 1 for all three families tested ($\sigma = (1,2), (1,1,2), (2,1)$), with 17–24% of gaps below $K - 1$.
- Alignment is not 1-Lipschitz along orbits: one-step alignment *gains* of +13, +15, +16 bits were observed for $\rho = -1, -19/11, -5$ respectively. (The mechanism: a step maps alignment-to-ρ to alignment-to-the-preimage-root $\rho^{-b} = (2^b\rho - 1)/3 \ne \rho$, so no per-step monotonicity exists.)

The correct statement available is supply-side, not speed-side: re-entries at depth g consume the band pool of Lemma 2 without replacement; nothing constrains *how soon* the next pool point is hit, only *how many times* it can ever be.

6 Position

The manuscript's Route-B core, the Carry Independence Conjecture — every $n_0 > 1$ eventually takes a letter $b \ge 3$ — is the extreme special case $\mathbb{P}[h \text{ odd}] \equiv 1$ of our joint calibration conjecture (its Cantor set $\mathcal{C}_\infty$ consists of the orbits all of whose +1-handoffs are odd forever). The balanced (bounded-discrepancy) core of $\mathcal{C}_\infty$ is already excluded unconditionally by repetition rigidity and the narrow-band corollary ($\sup_j |S_j| < 1.246$), which the manuscript leaves conditional. Its terminal wall — "distributional-to-pointwise promotion" — is our quenched calibration wall, reached independently. What we import, after verification: the universal repulsion mechanism (Theorem 1), the census lemma (Lemma 3, with the restored hypothesis), and the amortized constant $R = 0.0882$ (Proposition 1); what we contribute back: the exact ride cap, the supply bounds replacing the false return-time theorem, and the Lean-verified rigidity underlying the narrow-band exclusion.

References

[1] E. Y. Chang, *Exploring Collatz Dynamics with Human-LLM Collaboration*, arXiv:2603.11066v6 (2026).

[2] J. A. Janik, *Two Clocks: the* ± 1 *meters, depth supply, and unconditional escape bounds*, companion note (2026).

[3] J. A. Janik, *Narrow-band exclusion and the* (D, ε) *phase diagram*, companion note (2026).